

Magnetic Resonance-guided Fusion Ultrasound (MRgFUS) *for brain tumours*

Joan Gabrielle Basa | PHYS 483 | Literature Review



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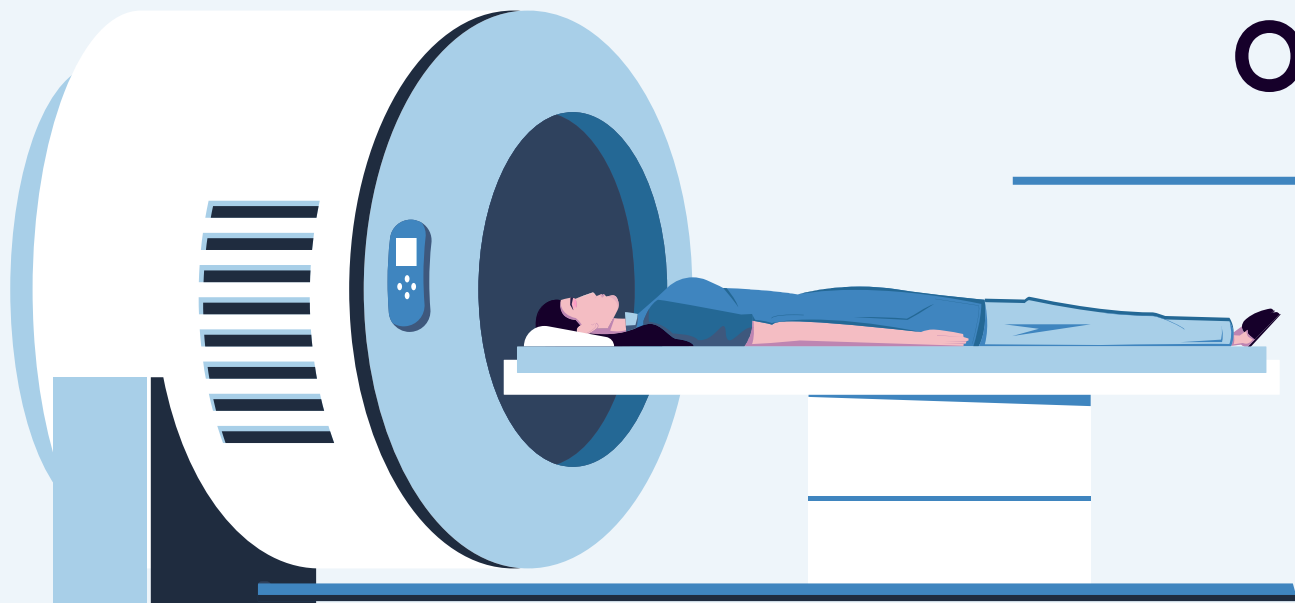
06

Future Directions



01

Overview



The basics

- **Definition**
 - Non-invasive therapy combining MRI guidance + Focused Ultrasound (FUS)
- **Key features**
 - Real-time imaging + temperature monitoring
 - No incisions, no ionizing radiation
- **How it works**
 - MRI locates target → FUS delivers energy → Tissue ablation/modulation



History



1942 (Lynn et al.)

focused sonication could
cause tissue destruction

1950s (Fry)

FUS could produce targeted
neural destruction

Early 1990s

integration of FUS and MRI into
a single system (MRgFUS)

Overview

Mechanisms

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Worldwide



① N. America

USA, Canada

② Europe

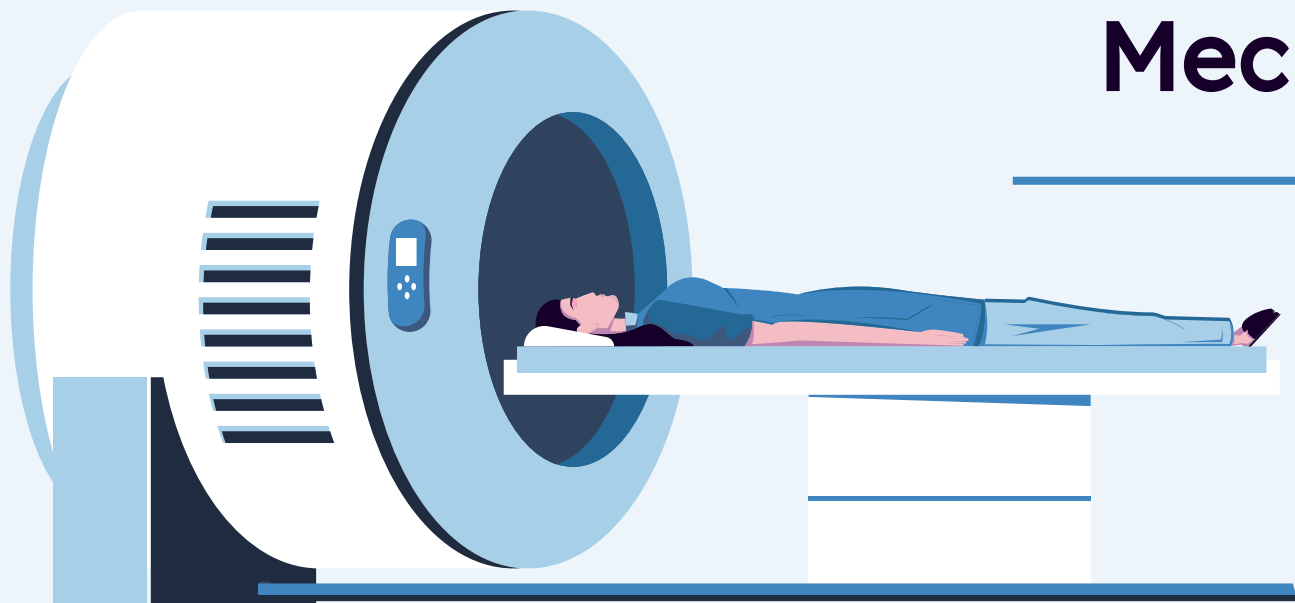
UK, Germany, Italy

③ Asia

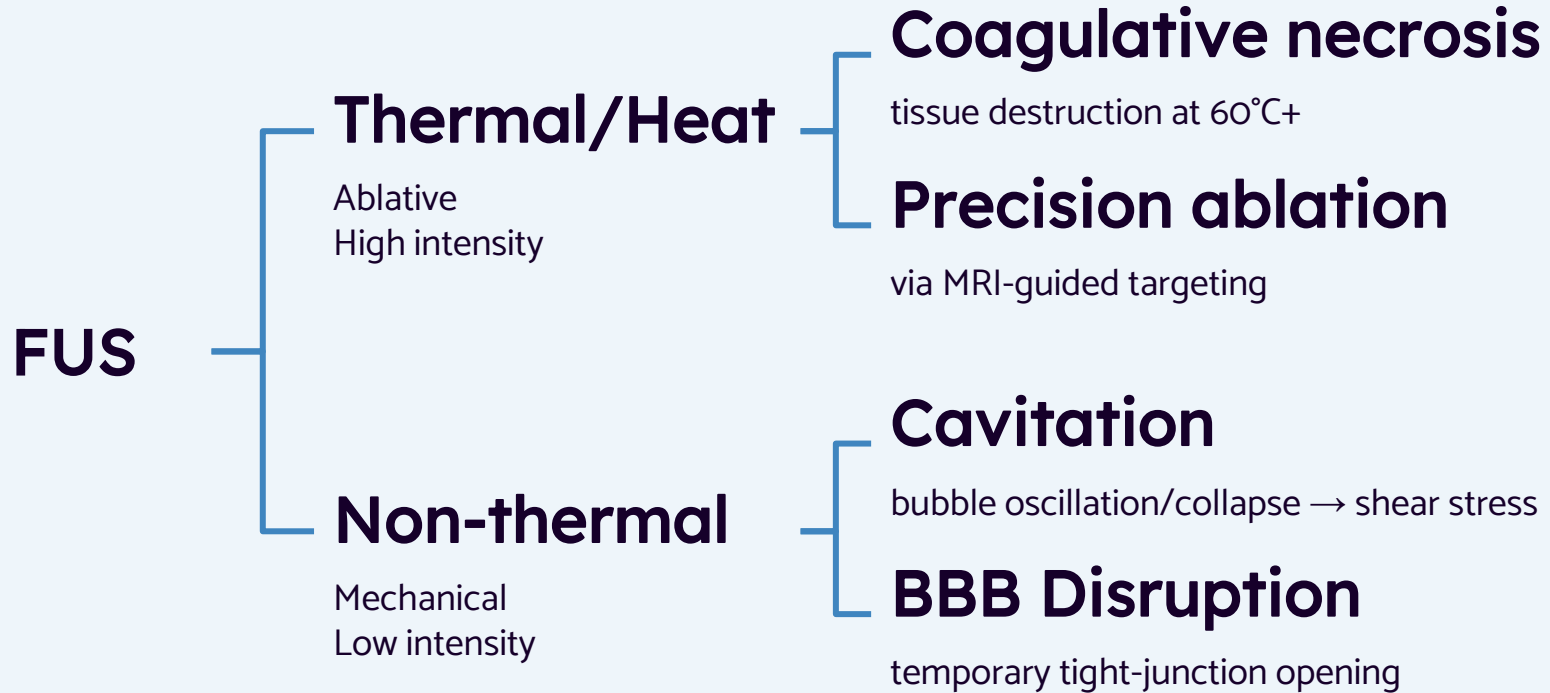
China, Japan

02

Mechanisms



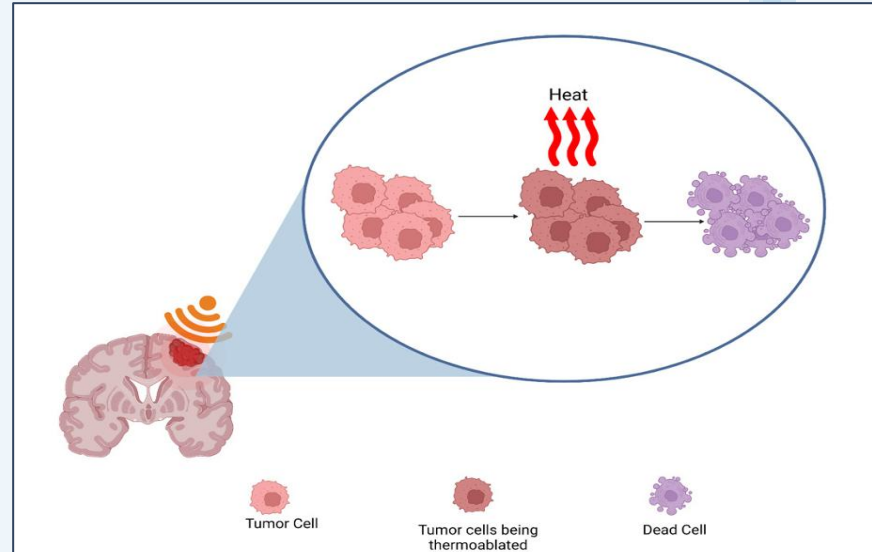
Mode of action (MoA)



Thermal effects

High intensity focused ultrasound (HIFU)

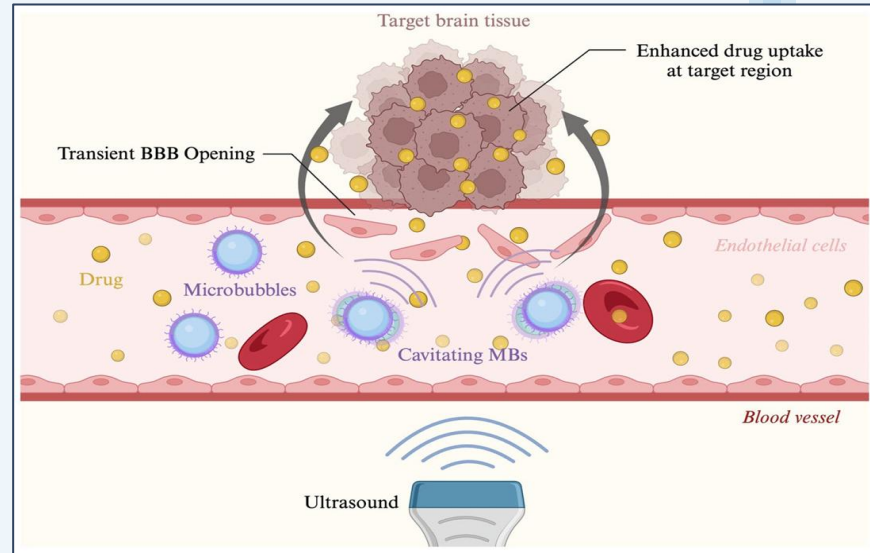
- Energy-to-heat conversion
 - Sound: 0.5 to 8 MHz
 - Heat: $>56^{\circ}\text{C}$
- Precision control parameters
 - Duration: 1 to 60s pulses
- Real-time MRI guidance
 - Anatomical targeting and temperature monitoring



Non-thermal effects

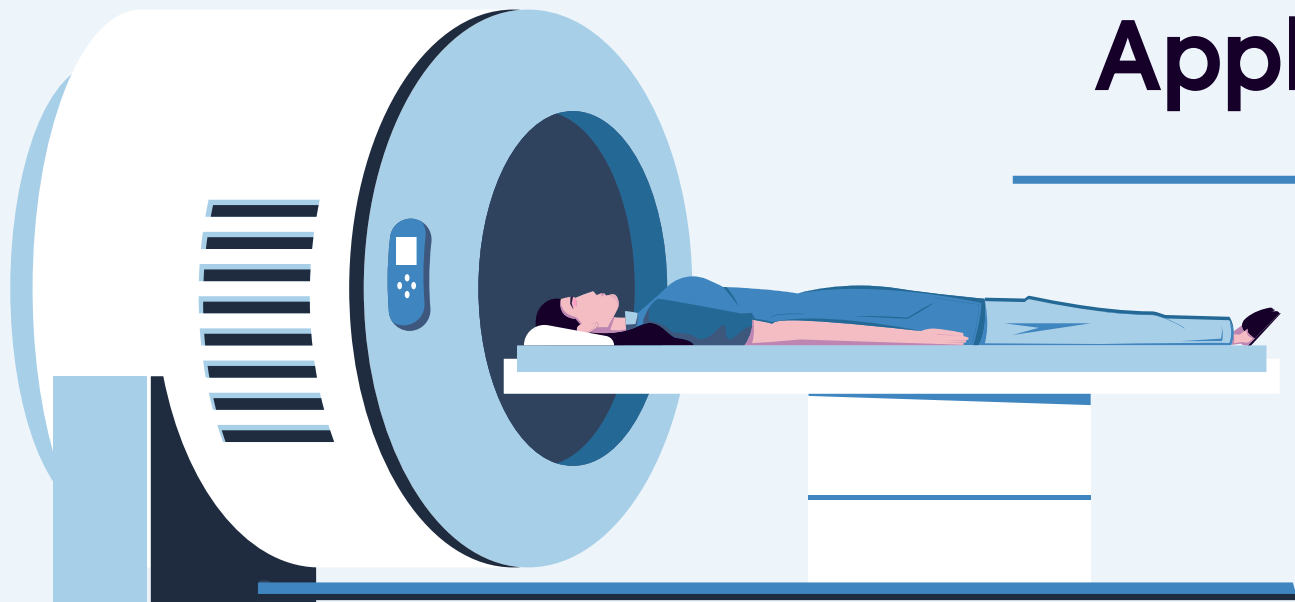
Low intensity focused ultrasound (LIFU)

- Cavitation
 - Stable cavitation (low pressure)
 - Inertial cavitation (high pressure)
- Precision control
 - 0.2 to 1.2 MHz
 - Microbubbles
- Real-time MRI guidance

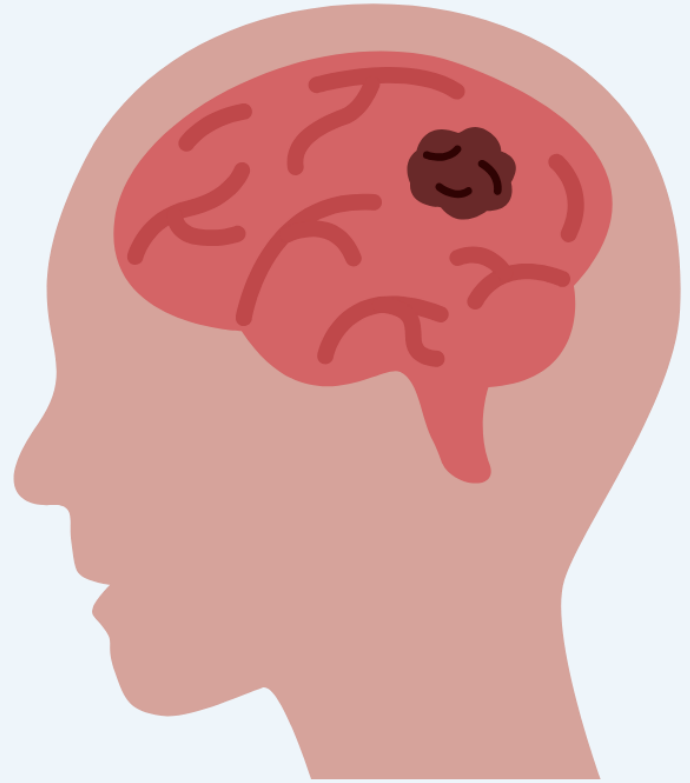


03

Applications



Brain tumours



Brain tumours (imaging + treatment)

- HIFU (Thermal Ablation) + LIFU (BBB Disruption)
 - **Types of brain tumors**
 - Glioblastoma multiforme (GBM)
 - Astrocytomas
 - Oligodendrogliomas
- brain metastases from breast cancer and other cancers
- other intracranial neoplasms

Procedure

	01	02	03	04
	Planning	Preparation	Sonification	Monitoring
LIFU (non-thermal)	MRI + CT	degassed water b/w scalp and transducer to reduce US attenuation	Phased US/MBs disrupt BBB via grid targeting	MRI guides targeting, thermometry controls heat
HIFU (thermal)		Real-time parameter adjustments	650 kHz continuous waves (2–10 s exposures) + phased arrays	MRI thermometry to monitor cavitation and thermal rise

An external view of a FUS helmet

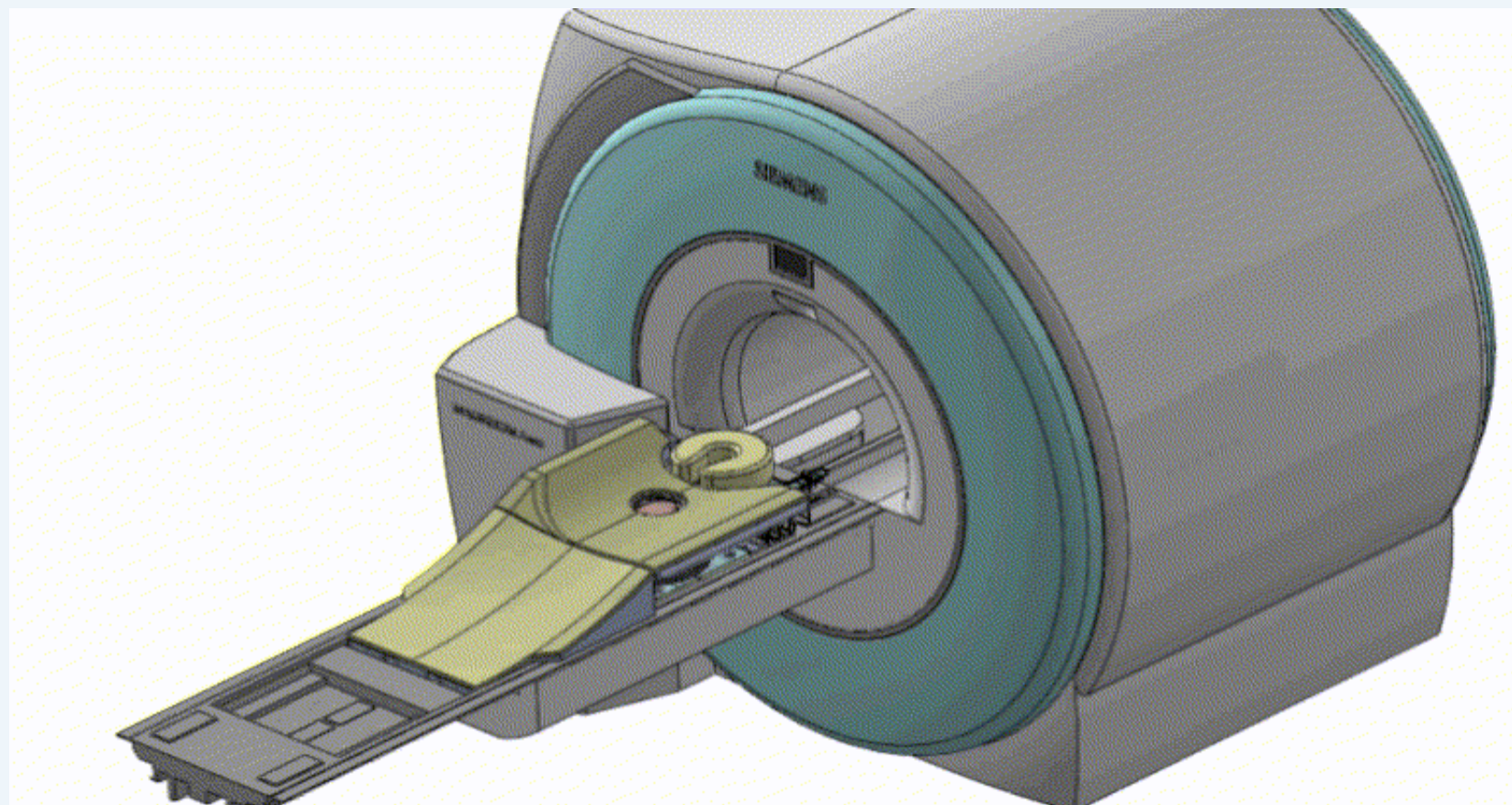


Other applications



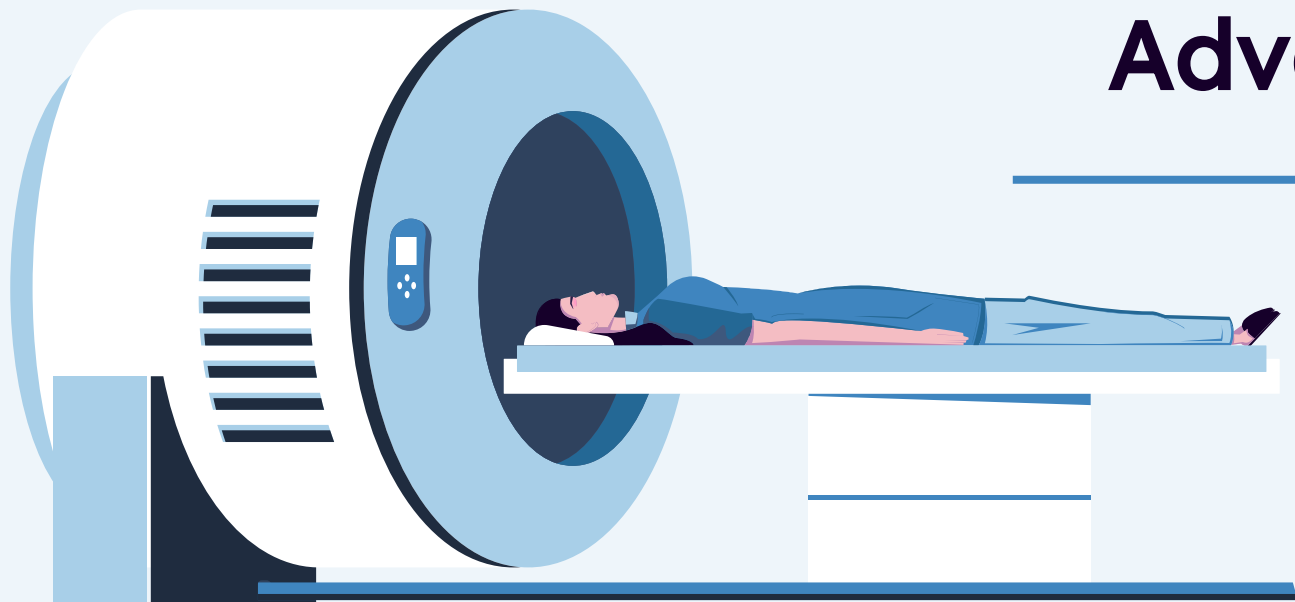
Other uses of MRgFUS

- **Essential tremors**
 - involuntary rhythmic contractions and relaxations of certain muscle groups
 - alternative to invasive neurosurgery → deep brain targets (FDA-approved in 2016)
- **Uterine fibroids**
 - benign smooth muscle tumors of the uterus
 - received FDA approval in the USA and in the UK
- **Breast tumors**
 - benign and malignant breast tumors



04

Advantages



Main advantages

01

Non-invasive

Does not require any incisions or opening of the skull to reach the target tumour volume

02

Repeatable

Non-toxic nature allows for repeated sessions over time, esp. for recurring tumours

03

BBB Disruption

Targeted and reversible disruption of BBB allows for enhanced delivery of therapeutic agents

Overview

Mechanisms

Applications

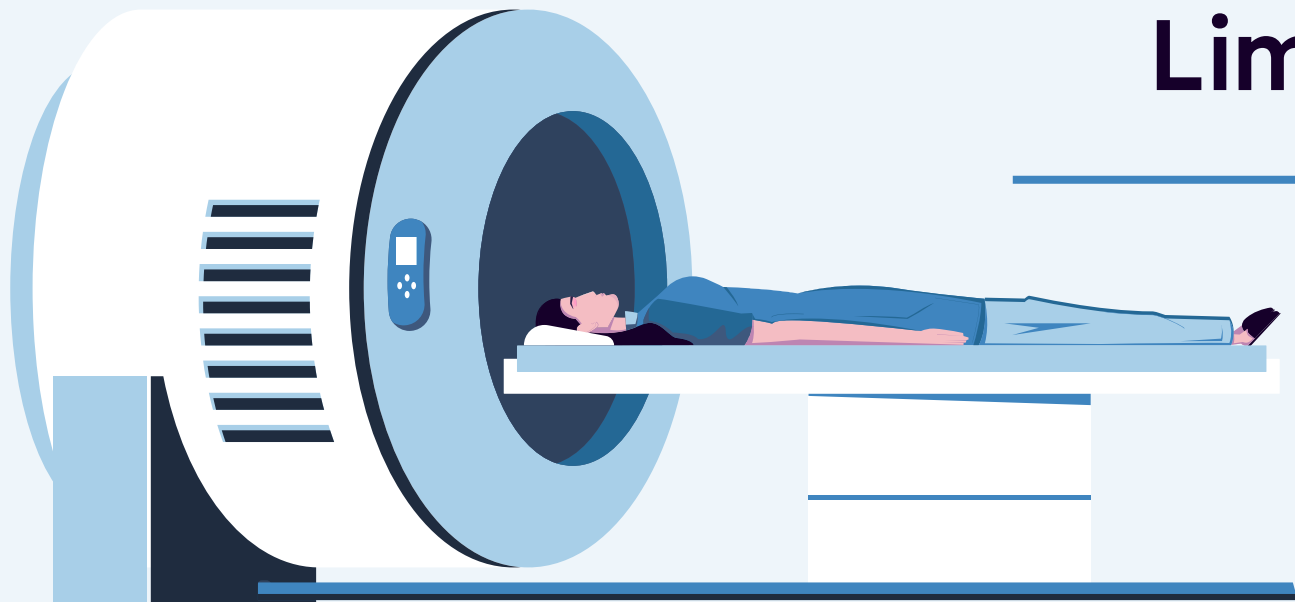
Advantages

Limitations

Future Directions

05

Limitations



Main limitations

01

Skull attenuation

Inhomogeneous thickness and density and the skull causes defocusing and heating

02

Non-selective

Both neoplastic and normal tissues located within the focal point are destroyed

03

Cost + Accessibility

High equipment maintenance cost and limited to specialized centers/institutions

Overview

Mechanisms

Applications

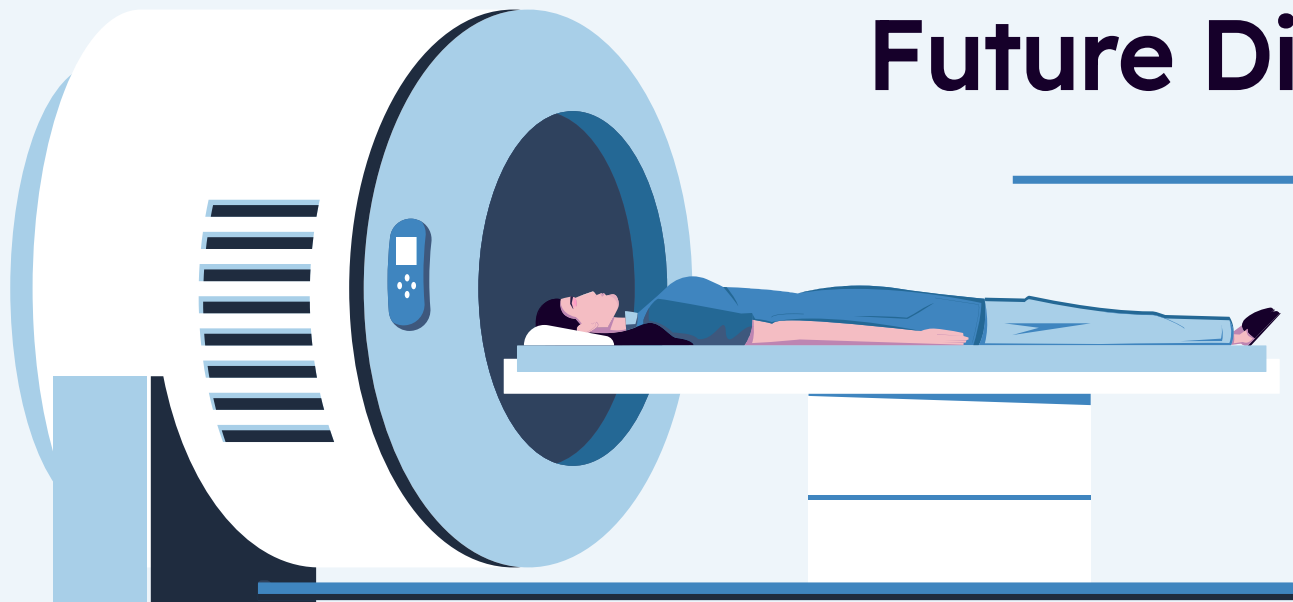
Advantages

Limitations

Future Directions

06

Future Directions



An illustration of a large, stylized red brain with white wavy lines representing neural pathways. A blue and red concentric circle target symbol is positioned on the right hemisphere of the brain. In the foreground, two stylized figures, an older man in a red shirt and an older woman in a blue shirt, are walking with a black metal walker. The background is light blue with several small, light blue and white plus signs scattered around.

Expanded Applications

Parkinson's disease
Alzheimer's disease

Overview

Mechanisms

Applications

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Future Directions



Non-thermal action

Neuromodulation
Immunomodulation

Overview

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References

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References

Mars

Despite being red, Mars is actually a cold place. It's full of iron oxide dust, which gives the planet its reddish cast, and it's made of basalt

Venus

Venus has a beautiful name and is the second planet from the Sun. It's terribly hot and its atmosphere is extremely poisonous

Jupiter

Jupiter is a gas giant and the biggest planet in the Solar System. It's the fourth-brightest object in the night sky and was named after a god

Saturn

Saturn is a gas giant and has several rings. It's composed mostly of hydrogen and helium. It's the sixth planet from the Sun



Reviewing concepts is a good idea



Mars

Despite being red, Mars is actually a cold place. It's full of iron oxide dust, which gives the planet its reddish cast

Venus

Venus has a beautiful name and is the second planet from the Sun. It's terribly hot, even hotter than Mercury

Neptune

Neptune is the farthest planet from the Sun. It's also the fourth-largest planet by diameter in the Solar System

Earth

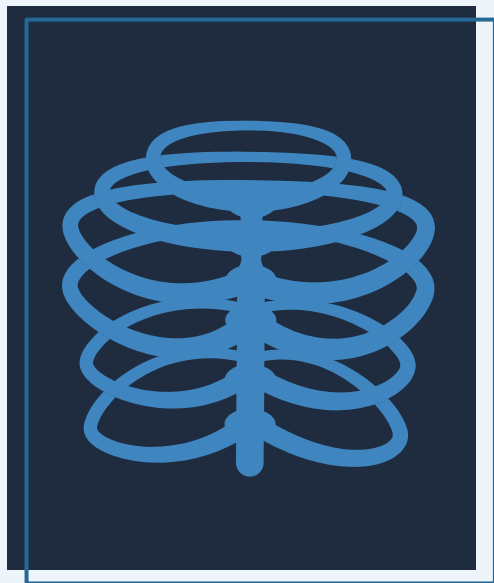
Earth is the third planet from the Sun and the only one that harbors life in the Solar System. We all live on this planet

Saturn

Saturn is a gas giant and has several rings. It's composed of hydrogen and helium. It's the sixth planet from the Sun

Jupiter

Jupiter is a gas giant and the biggest planet in the Solar System. It's the fourth-brightest object in the night sky



“This is a quote, words
full of wisdom that
someone important said
and can make the reader
get inspired”

—Someone Famous

Is this infographic useful for you?

Mercury

Study 1

Mercury is a small planet

Venus

Study 2

Venus has a beautiful name

Earth

Study 3

Earth is the planet with life

Neptune

Study 4

Neptune is far away from us

Mars

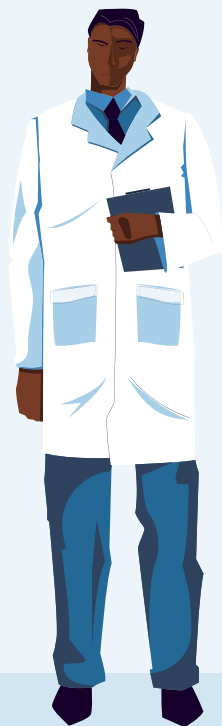
Study 5

Mars was named after a god

Jupiter

Study 6

Jupiter is a huge gas giant



98,300,000

Big numbers catch your audience's attention



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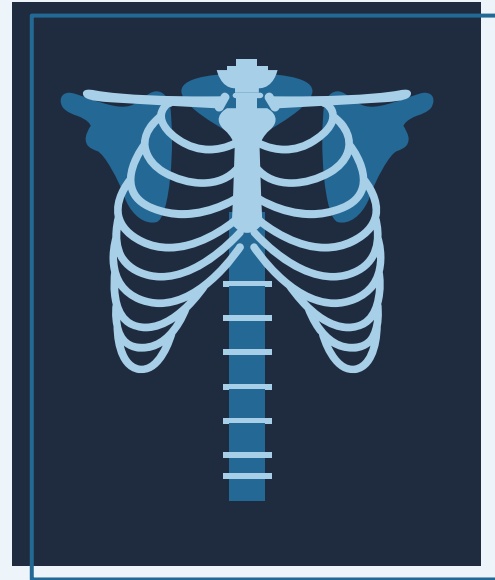
Jupiter's rotation period

333,000

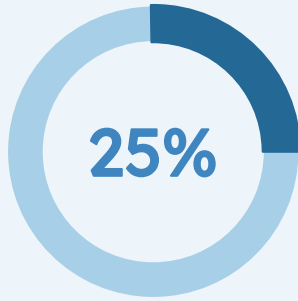
The Sun's mass compared to Earth's

386,000 km

Distance between Earth and the Moon

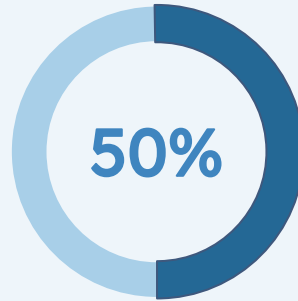


Let's use some percentages



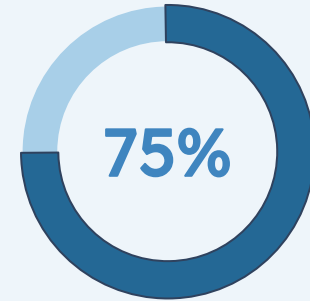
Mercury

Mercury is the closest planet to the Sun and the smallest in the Solar System



Venus

Venus has a beautiful name and is the second planet from the Sun



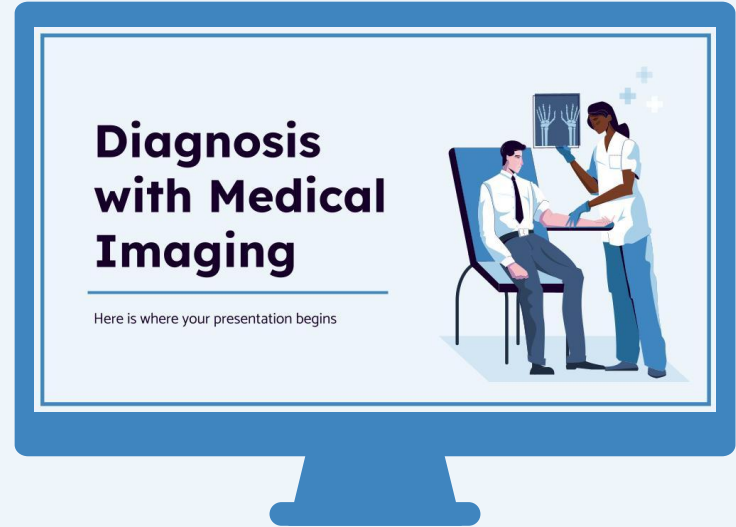
Mars

Despite being red, Mars is actually a cold place. It's full of iron oxide dust



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Data about two tests

Test 1



Jupiter is a gas giant

Saturn has rings

Test 2



Pluto is a dwarf planet

Earth harbors life

Mercury

Mercury is the closest planet to the Sun

Saturn

It's composed of hydrogen and helium

Jupiter

It's the biggest planet in the Solar System

Mars

Mars is actually a very cold planet



A timeline always works well

Earth

Earth is the third planet from the Sun

Moon

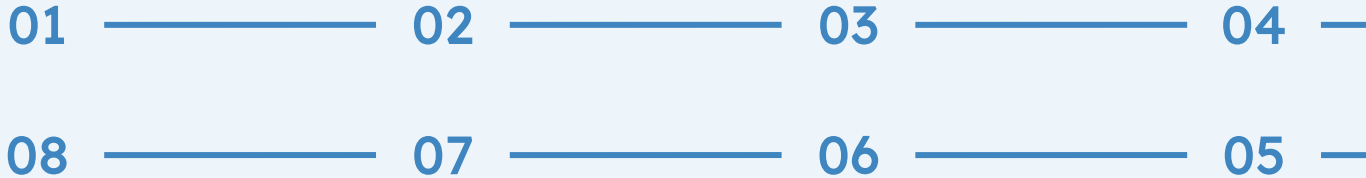
The Moon is Earth's only natural satellite

Mercury

Mercury is the closest planet to the Sun

Saturn

Saturn was named after a Roman god



Neptune

Neptune is the farthest planet from the Sun

Mars

Despite being red, Mars is a cold place

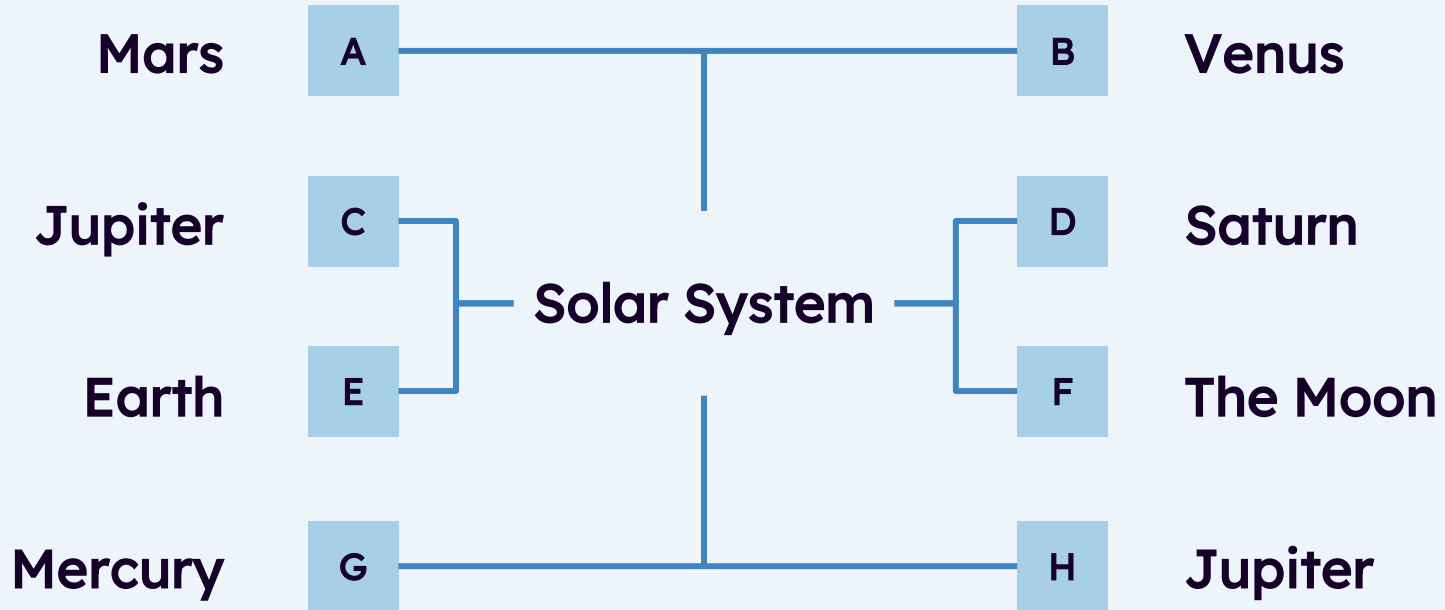
Jupiter

Jupiter was named after a Roman god

Ceres

Ceres is located in the main asteroid belt

Infographics are useful



Use tables to represent data

	Team A	Team B	Team C	Team D	Team E	Team F
Mercury	XX	XX	XX	XX	XX	XX
Mars	XX	XX	XX	XX	XX	XX
Saturn	XX	XX	XX	XX	XX	XX
Venus	XX	XX	XX	XX	XX	XX
Jupiter	XX	XX	XX	XX	XX	XX
Earth	XX	XX	XX	XX	XX	XX
Moon	XX	XX	XX	XX	XX	XX

You can use this graph

● 20XX

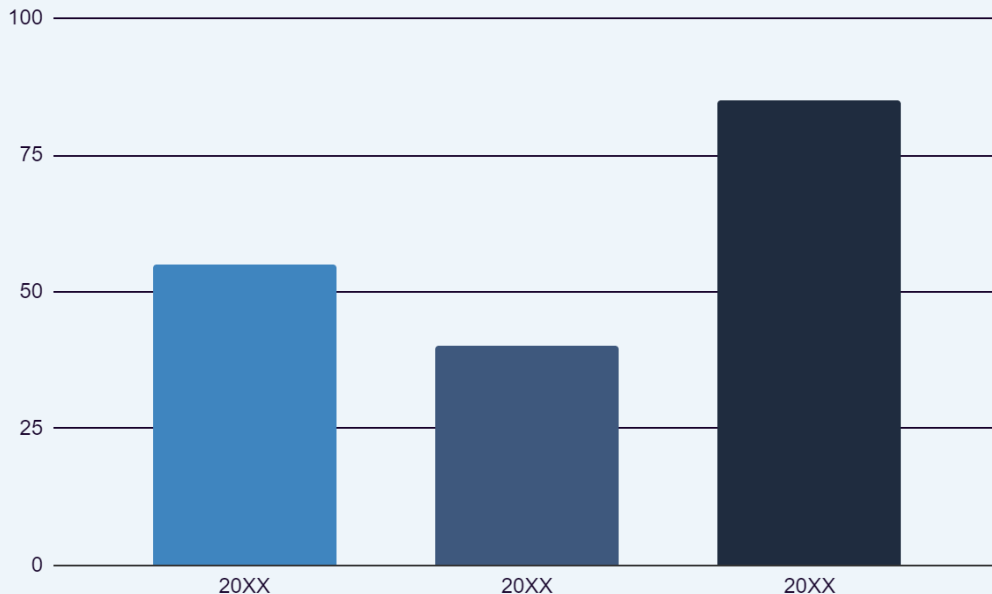
Mars is very cold

● 20XX

Mercury is small

● 20XX

Venus is very hot



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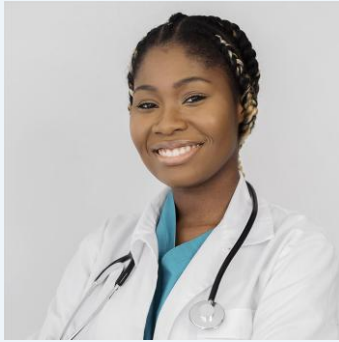
One last diagram

Study of the image

Saturn was named after the Roman god
of wealth and agriculture



Our team



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person here



Tom Harris

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person here



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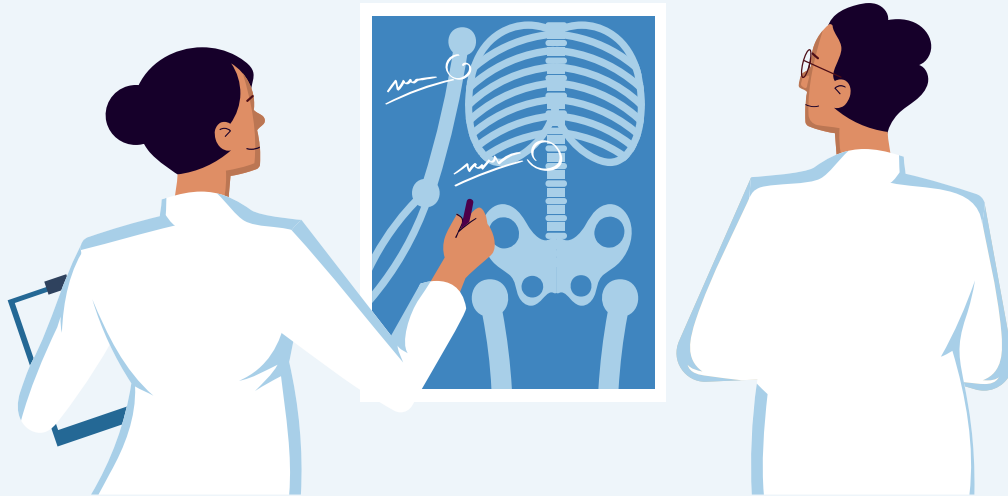


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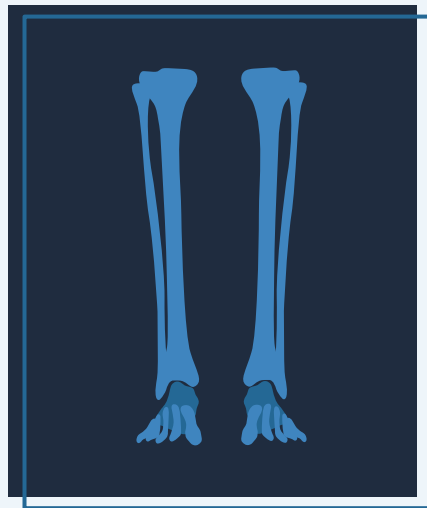
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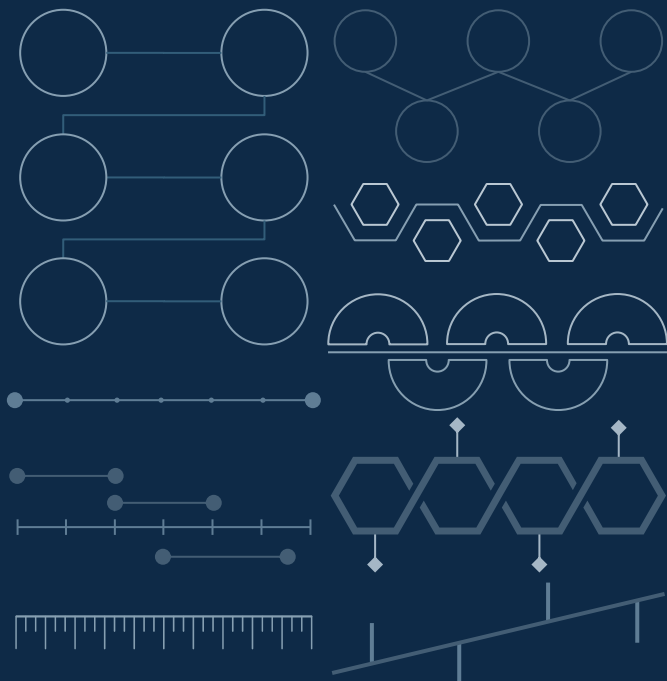
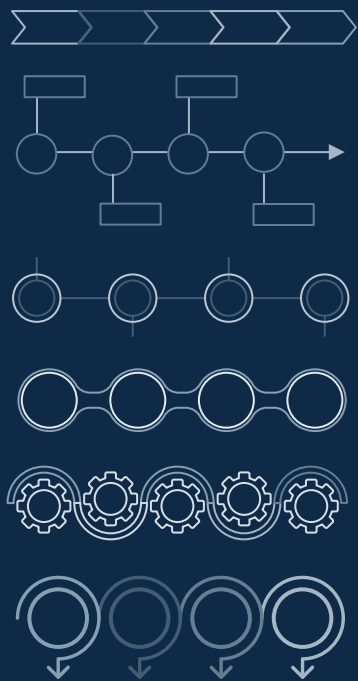
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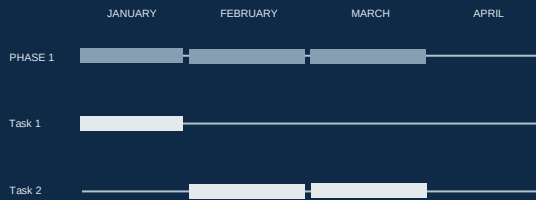
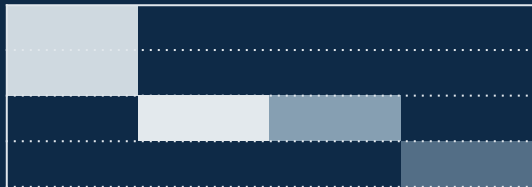
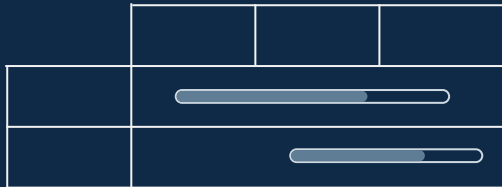
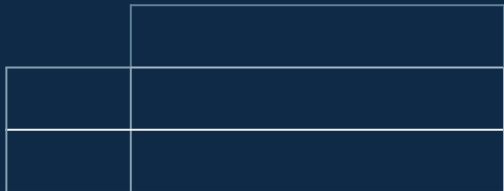
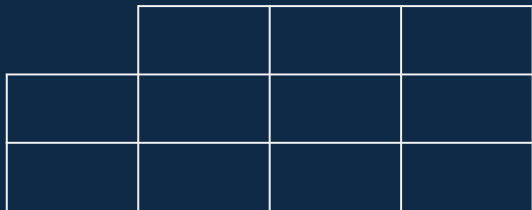
Use our editable graphic resources...

You can easily **resize** these resources without losing quality. To **change the color**, just ungroup the resource and click on the object you want to change. Then, click on the paint bucket and select the color you want. Group the resource again when you're done. You can also look for more **infographics** on [Slidesgo](#).

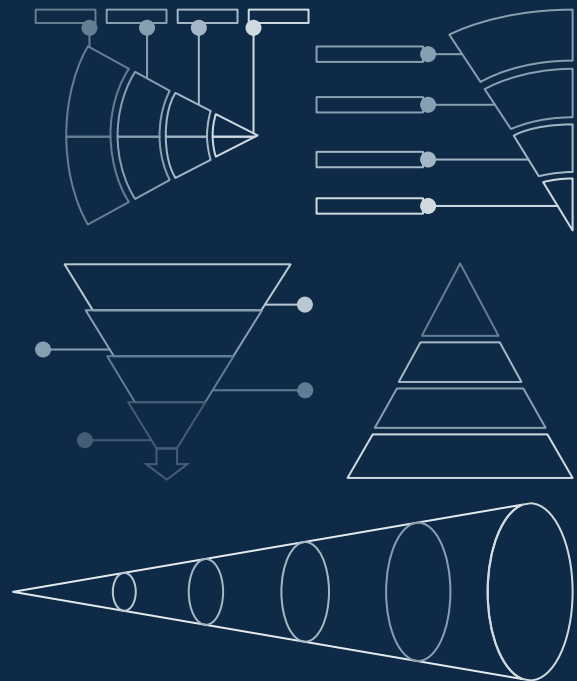
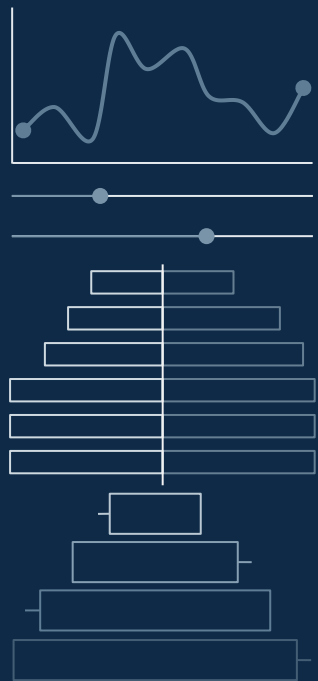
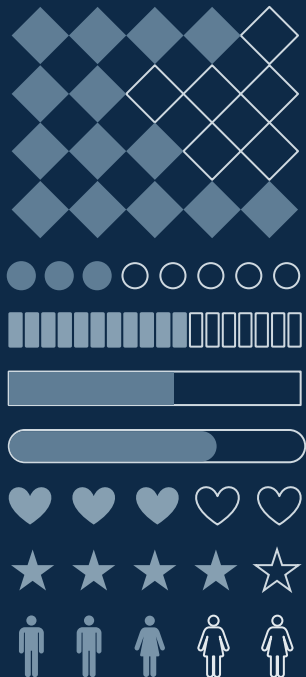
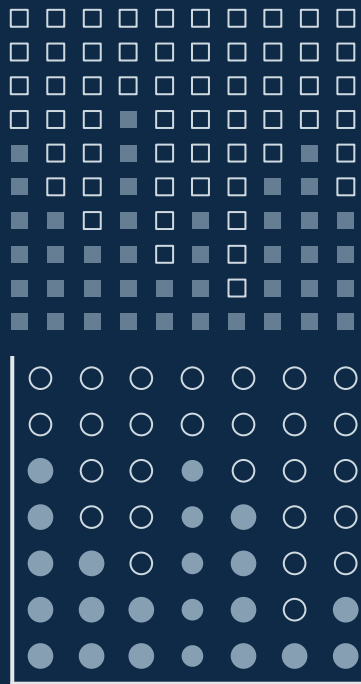












...and our sets of editable icons

You can **resize** these icons without losing quality.

You can **change the stroke and fill color**; just select the icon and click on the **paint bucket/pen**.

In Google Slides, you can also use **Flaticon's extension**, allowing you to customize and add even more icons.



Educational Icons



Medical Icons



Business Icons



Teamwork Icons



Help & Support Icons



Avatar Icons



Creative Process Icons



Performing Arts Icons



Nature Icons



SEO & Marketing Icons



